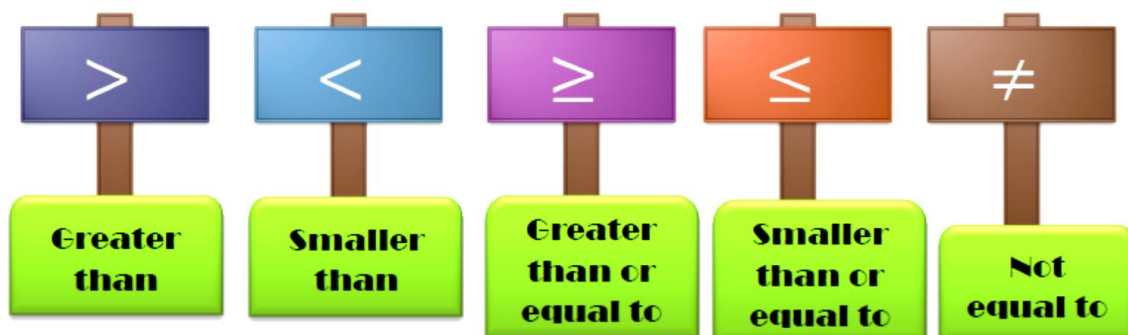


Answer

LESSON 3 : INEQUALITY



Checkpoint 1

$$\begin{aligned} 2x - 1 &= 5 \\ 2x &= 6 \\ x &= 3 \end{aligned}$$

$$\begin{aligned} 2x - 1 &> 5 \\ 2x &> 6 \\ x &> 3 \end{aligned}$$

$$\begin{aligned} x + 3 &< 5 \\ x &< 5 - 3 \\ x &< 2 \end{aligned}$$

$$\begin{aligned} x - 2 &> 10 \\ x &> 10 + 2 \\ x &> 12 \end{aligned}$$

$$\begin{aligned} 5 &< 2x - 1 \\ 6 &< 2x \\ 3 &< x \\ x &> 3 \end{aligned}$$

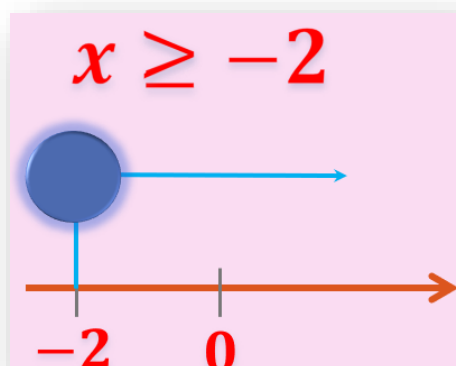
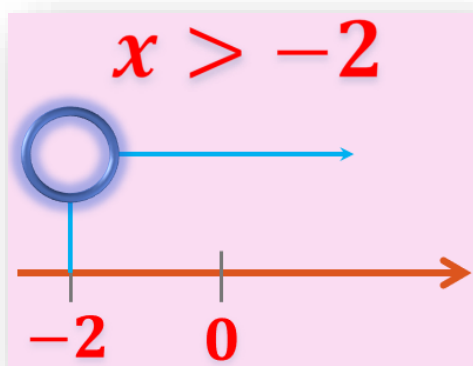


Make subject

主角放左邊

唔好冇大有細

Checkpoint 2



Let's get started!

Solve the following inequalities graphically.

1. $x + 3 < 7$

$$x < 4$$

2. $x - 3 > 7$

$$x > 10$$

3. $x + 6 \leq 10$

$$x \leq 4$$

4. $7 < x - 15$

$$x > 22$$

5. $11 \geq 3 + x$

$$x \leq 8$$

6. $10 + x < 27$

$$x < 17$$

7. $10 < 15 - x$

$$x < 5$$

8. $209 \leq 300 - x$

$$x \leq 91$$

9. $287 \geq 106 + x$

$$x \leq 181$$

10. $256 - x < 96$

$$x > 160$$

Checkpoint 3

$$\begin{aligned}5x &\leq 15 \\ x &\leq \frac{15}{5} \\ x &\leq 3\end{aligned}$$

$$\begin{aligned}\frac{x}{4} &\geq 2 \\ x &\geq 2 \times 4 \\ x &\geq 8\end{aligned}$$

You have learnt more!

Solve the following inequalities graphically.

11. $3x < 15$

$$x < 5$$

12. $2x < 86$

$$x < 43$$

13. $\frac{x}{8} < 9$

$$x < 72$$

14. $\frac{x}{7} \leq 13$

$$x \leq 91$$

15. $15 < \frac{x}{6}$

$$x > 90$$

16. $32 \geq \frac{x}{3}$

$$x \leq 96$$

17. $\frac{24x}{5} \geq 48$

$$x \geq 10$$

18. $258 < \frac{3x}{2}$

$$x > 172$$

19. $9 < \frac{3x}{35}$

$$x > 105$$

20. $\frac{3x}{7} < 8$

$$x < \frac{56}{3}$$

21. $\frac{5x}{18} < 10$

$$x < 36$$

22. $\frac{216x}{9} \leq 24$

$$x \leq 1$$

Checkpoint 4

$$\begin{aligned} -\frac{x}{4} &= 2 \\ -x &= 2 \times 4 \\ -x &= 8 \\ -8 &= x \\ x &= -8 \end{aligned}$$

$$\begin{aligned} -\frac{x}{4} &\geq 2 \\ -x &\geq 2 \times 4 \\ -x &\geq 8 \\ \text{開口} \quad \text{尖頭} \\ -8 &\geq x \\ x &\leq -8 \\ \text{尖頭} \quad \text{開口} \end{aligned}$$

$$-x < 4$$

$$x > -4$$



乘除負數

大細調轉

Keep Fighting!

23. $-x + 3 > 7$

$$x < -4$$

24. $-2x + 5 < 17$

$$x > -6$$

25. $x + 2 \geq 4 + 3x$

$$x \leq -1$$

26. $13 \leq 9x - 5$

$$x \geq 2$$

27. $-4x + 7 > 5$

$$x < \frac{1}{2}$$

28. $2x + 3 < 10x - 5$

$$x > 1$$

29. $10 - 2x \geq x + 1$

$$x \leq 3$$

30. $5 - 3x \leq -1 + 3x$

$$x \geq 1$$

31. $x + 4 > 4x - 8$

$$x < 4$$

32. $-30 + 12x < 42$

$$x < 6$$

33. $4(x - 1) \geq 7x - 10$

$$x \leq 2$$

34. $6(3 - x) \leq 4(-2x + 3)$

$$x \leq -3$$

PRE S3 MATHS L3 - INEQUALITY

35. $2x + 3 > 5(2x - 1)$

$$x < 1$$

36. $6(3 - x) < 4(-2x + 3)$

$$x < -3$$

37. $12 - 2x \geq x + 3$

$$x \leq 3$$

38. $4 - 2x \leq -1 + 2x$

$$x \geq 1$$

39. $x + 4 > 4x - 8$

$$x < 4$$

40. $5x - 1 < 3x - 7$

$$x < -3$$

41. $4x - 12 \geq 10 - 7x$

$$x \geq 2$$

42. $13 - 5x \leq 23 - 7x$

$$x \leq 5$$

43. $-6(5 - 2x) > 42$

$$x > 6$$

44. $5(x - 8) < 10$

$$x < 10$$

45. $\frac{1}{5}(-4 - x) \geq 8$

$$x \leq -44$$

46. $\frac{1}{3}(4x - 1) \leq 1$

$$x \leq 1$$

Give it a try

$$\begin{aligned}3(x+2) - 5(x+4) &> 0 \\3x + 6 - 5x - 20 &> 0 \\-2x - 14 &> 0 \\-2x &> 14 \\-14 &> 2x \\-7 &> x \\x &< -7\end{aligned}$$

$$\begin{aligned}2x + 1 &< \frac{4x + 1}{3} \\3(2x + 1) &< 4x + 1 \\6x + 3 &< 4x + 1 \\6x - 4x &< +1 - 3 \\2x &< -2 \\x &< -1\end{aligned}$$



一邊有分母 n ,
其他 terms 全部 $\times \frac{n}{n}$

Challenging!!

Case 1

$$\frac{2 - 3x}{4} \leq 1 - x$$

$$\frac{2 - 3x}{4} \leq (1 - x) \times \frac{4}{4}$$

$$2 - 3x \leq 4(1 - x)$$

$$2 - 3x \leq 4 - 4x$$

$$-3x + 4x \leq 4 - 2$$

$$x \leq 2$$

Case 2

$$\frac{5x}{3} - 2 < \frac{10x}{9} + 3$$

$$\frac{5x}{3} - \frac{10x}{9} < 3 + 2$$

$$\frac{5x}{3} \times \frac{9}{9} - \frac{10x}{9} \times \frac{3}{3} < 5 \times \frac{27}{27}$$

$$45x - 30x < 135$$

$$x < 9$$

$$\frac{9}{9} \times \frac{3}{3}$$



Final Boss!

1. $\frac{x+1}{2} \geq \frac{9x+5}{19}$

$$\begin{aligned}\frac{x+1}{2} &\geq \frac{9x+5}{19} \\ 38\left(\frac{x+1}{2}\right) &\geq 38\left(\frac{9x+5}{19}\right) \\ 19(x+1) &\geq 2(9x+5) \\ 19x+19 &\geq 18x+10 \\ 19x-18x &\geq 10-19 \\ \underline{\underline{x &\geq -9}}\end{aligned}$$

2. $\frac{3x+8}{2} \geq \frac{3-x}{5}$

$$\begin{aligned}\frac{3x+8}{2} &\geq \frac{3-x}{5} \\ 10\left(\frac{3x+8}{2}\right) &\geq 10\left(\frac{3-x}{5}\right) \\ 5(3x+8) &\geq 2(3-x) \\ 15x+40 &\geq 6-2x \\ 15x+2x &\geq 6-40 \\ 17x &\geq -34 \\ x &\geq \frac{-34}{17} \\ \underline{\underline{x &\geq -2}}\end{aligned}$$

3. $\frac{6-x}{11} - \frac{1-x}{4} < 0$

$$\begin{aligned}\frac{6-x}{11} - \frac{1-x}{4} &< 0 \\ 4(6-x) - 11(1-x) &< 0 \\ 24-4x-11+11x &< 0 \\ 7x &< -13 \\ x &< \underline{\underline{-\frac{13}{7}}}\end{aligned}$$

4. $\frac{5x-3}{4} - \frac{7x-25}{3} \leq 0$

$$\begin{aligned}\frac{5x-3}{4} - \frac{7x-25}{3} &\leq 0 \\ 3(5x-3) - 4(7x-25) &\leq 0 \\ 15x-9-28x+100 &\leq 0 \\ -13x &\leq -91 \\ x &\geq \frac{-91}{-13} \\ \underline{\underline{x &\geq 7}}\end{aligned}$$

5. $x+16 > \frac{2x}{13} + 5$

$$\begin{aligned}x+16 &> \frac{2x}{13} + 5 \\ x - \frac{2x}{13} &> 5-16 \\ \frac{11x}{13} &> -11 \\ x &> -11 \times \frac{13}{11} \\ \underline{\underline{x &> -13}}\end{aligned}$$

6. $\frac{5x}{3} - 2 < \frac{10x}{9} + 3$

$$\begin{aligned}\frac{5x}{3} - 2 &< \frac{10x}{9} + 3 \\ \frac{5x}{3} - \frac{10x}{9} &< 3+2 \\ \frac{5x}{9} &< 5 \\ x &< 5 \times \frac{9}{5} \\ \underline{\underline{x &< 9}}\end{aligned}$$

7. $4x+16+\frac{x}{2} < 13$

$$\begin{aligned}4(x+4) + \frac{x}{2} &< 13 \\ 4x+16+\frac{x}{2} &< 13 \\ \frac{9}{2}x &< -3 \\ x &< \underline{\underline{-\frac{2}{3}}}\end{aligned}$$

8. $7-4x > \frac{x+1}{3}$

$$\begin{aligned}21-12x &> x+1 \\ 20 &> 13x \\ x &< \underline{\underline{\frac{20}{13}}}\end{aligned}$$

PRE S3 MATHS L3 - INEQUALITY

9. $1 + \frac{x}{7} + \frac{x-3}{2} \geq 3(x+1)$

$$\begin{aligned} 1 + \frac{x}{7} + \frac{x-3}{2} &\geq 3(x+1) \\ 14 + 2x + 7x - 21 &\geq 42x + 42 \\ 14 - 21 - 42 &\geq 42x - 2x - 7x \\ -49 &\geq 33x \\ \frac{-49}{33} &\geq x \\ x &\leq \frac{-49}{33} \end{aligned}$$

10. $5x - \frac{2x-9}{5} \leq x - 3$

$$\begin{aligned} 5x - \frac{2x-9}{5} &\leq x - 3 \\ 25x - 2x + 9 &\leq 5x - 15 \\ 25x - 2x - 5x &\leq -15 - 9 \\ 18x &\leq -24 \\ x &\leq -\frac{4}{3} \end{aligned}$$

11. $3 - \frac{x+1}{2} \geq 5 - \frac{x+2}{3}$

$$\begin{aligned} 3 - \frac{x+1}{2} &\geq 5 - \frac{x+2}{3} \\ 18 - 3x - 3 &\geq 30 - 2x - 4 \\ 18 - 3 - 30 + 4 &\geq -2x + 3x \\ -11 &\geq x \\ x &\leq -11 \end{aligned}$$

12. $\frac{1-2x}{3} - \frac{3-4x}{5} > -2$

$$\begin{aligned} \frac{1-2x}{3} - \frac{3-4x}{5} &> -2 \\ (15)\left(\frac{1-2x}{3} - \frac{3-4x}{5}\right) &> (15)(-2) \\ (5)(1-2x) - (3)(3-4x) &> -30 \\ 5 - 10x - 9 + 12x &> -30 \\ 2x - 4 &> -30 \\ 2x &> -26 \\ x &> -13 \end{aligned}$$

13. $\frac{-3x+5}{2} \geq \frac{11-x}{3}$

$$\begin{aligned} \frac{-3x+5}{2} &\geq \frac{11-x}{3} \\ 6\left(\frac{-3x+5}{2}\right) &\geq 6\left(\frac{11-x}{3}\right) \\ 3(-3x+5) &\geq 2(11-x) \\ -9x+15 &\geq 22-2x \\ 15-22 &\geq -2x+9x \\ -7 &\geq 7x \\ -1 &\geq x \\ x &\leq -1 \end{aligned}$$

14. $\frac{x}{6} + 1 \leq \frac{5}{6} - x$

$$\begin{aligned} \frac{x}{6} + 1 &\leq \frac{5}{6} - x \\ \frac{x}{6} + x &\leq \frac{5}{6} - 1 \\ \frac{7x}{6} &\leq -\frac{1}{6} \\ 7x &\leq -1 \\ x &\leq -\frac{1}{7} \end{aligned}$$